

Machine Learning for Customer Churn Prediction

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1 Executive Summary

Customer Churn is a problem about customer discontinue a company's services or products. Machine learning techniques can be applied with purpose of predict if a customer has probability of be churn, this information can help the company to take decisions for not lost clients. This work search explore different machine learning techniques to compare and get the best approach to solve this problem.

2 Dataset Download and Integrity Verification

2.1 Dataset Source and Acquisition

The Telco Customer Churn dataset was obtained from the public Kaggle page published by BlastChar and identified as originating from IBM Sample Data Sets. To reproduce the setup, download the complete `WA_Fn-UseC_-Telco-Customer-Churn.csv` file from <https://www.kaggle.com/datasets/blastchar/telco-customer-churn> and place it at `data/WA_Fn-UseC_-Telco-Customer-Churn.csv`. The Kaggle page does not specify a standard open-source license; this project uses the data for academic learning and analysis and acknowledges its source and original authors.

2.2 Dataset Characteristics

The CSV is 977,501 bytes (approximately 977.5 KB) and contains 7,043 customer records and 21 columns: 20 customer and service features, plus the binary `Churn` target. It has 1,869 `Yes` and 5,174 `No` records, indicating class imbalance.

2.3 Integrity Verification

The provider does not supply an official checksum. To calculate the SHA-256 hash of the local project copy, run `pythonsrc/w1_supervised_learning/dataset_info/checksum.py` from the repository root. The script reads the CSV under `data/` and prints its checksum:

```
88BE4B93FBE0CC83421AF1C503794C97C342ECA914C1576DB7C276E61D61358A
```

This value is a local reference because Kaggle does not provide an official checksum for comparison. To generate a structured dataset report, run `pythonsrc/w1_supervised_learning/dataset_info/dataset_info.py` from the repository root. It prints JSON with the file size, row and column counts, samples per **Churn** class, each column's data type and missing-value count, and numeric feature ranges. Both scripts use the Telco CSV by default; a different file can be supplied with `--csv`.

3 Exploratory Data Analysis (EDA)

This section explore the data, explain what represents each feature and respective type, the proportion in the values of numerical and categorical features, the relation between each feature with the target and visual representation of statistics obtained.

All process in this section can be explore in EDA notebook.

Feature	Description	Data type
customerID	Unique value that identify a customer	str
gender	Determines whether customer is Male or Female	str
SeniorCitizen	Determines whether a person is citizen or not	int64
Partner	Determines whether person has partner	str
Dependents	Determines whether customer has a dependent	str
tenure	Number of months has stayed with the company	int64
PhoneService	Whether customer has phone service	str
MultipleLines	Whether the customer has multiple lines, or one, or does not have phone service	str
InternetService	Internet service provider (Fiber optic, DSL or no)	str
OnlineSecurity	Whether customer has online security, or not, or does not have internet service	str
OnlineBackup	Whether customer has online backup, or not, or does not have internet service	str
DeviceProtection	Whether customer has device protection, or not, or does not have internet service	str
TechSupport	Whether customer has tech support, or not, or does not have internet service	str
StreamingTV	Whether customer has streaming tv, or not, or does not have internet service	str
StreamingMovies	Whether customer has streaming movies, or not, or does not have internet service	str
Contract	Contract term duration (Month-to-month, One year, Two year)	str
PaperlessBilling	Whether customer has paperless billing or not	str
PaymentMethod	Payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card)	str
MonthlyCharges	Value paid by customer monthly	float64
TotalCharges	Total value paid by customer	str
Churn	Whether the customer churned or not, stopped using the service last month.	str

Table 1: Dataset features description

3.1 Target Distribution

Of the 7,043 customers, 5,174 (73.46%) stayed and 1,869 (26.54%) churned. The majority-to-minority ratio is 2.77:1, so the target is moderately imbalanced. The class counts are summarized in Table ?? and visualized as a bar chart in the EDA notebook.

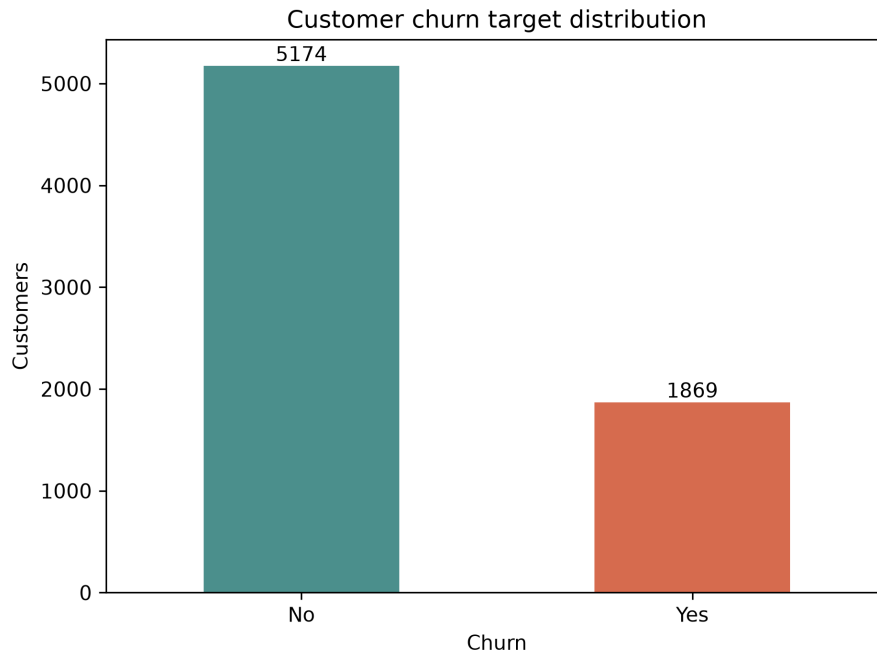


Figure 1: Distribution of customers by churn status.

3.2 Feature Statistics

Table 2 summarizes the non-binary numeric features: customer tenure and charges. The binary **SeniorCitizen** feature is excluded from these measurements.

Feature	N	Mean	SD	Min	Max
tenure (months)	7,043	32.37	24.56	0	72
MonthlyCharges	7,043	64.76	30.09	18.25	118.75
TotalCharges	7,032	2,283.30	2,266.77	18.80	8,684.80

Table 2: Summary statistics for numeric features.

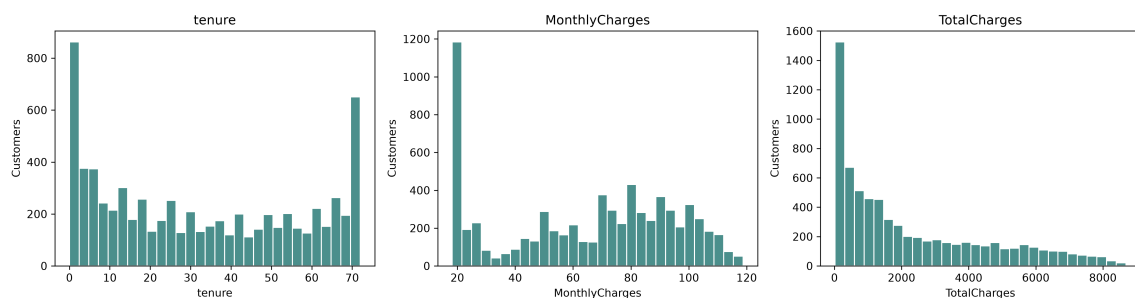


Figure 2: Histograms of tenure, monthly charges, and total charges.

3.3 Categorical Features

The categorical features describe customer demographics and household status, phone and internet services, service add-ons, streaming, contract and billing arrangements, and payment method. Each bar shows the percentage of all 7,043 customers in that category. The unique identifier `customerID` and target `Churn` are excluded here; the target distribution is shown separately in Figure 1.

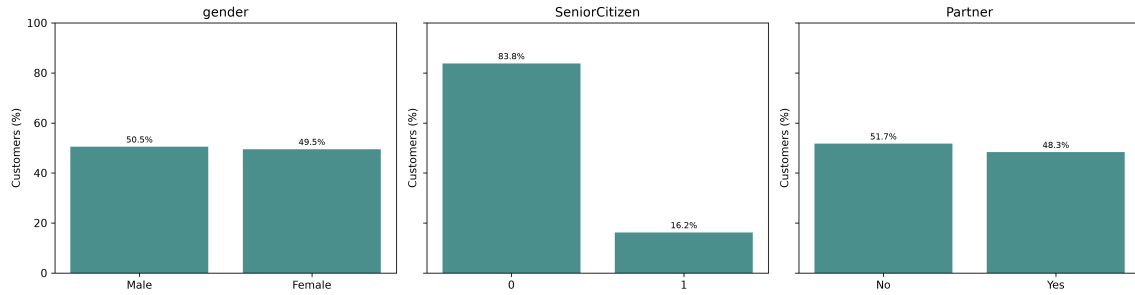


Figure 3: Distributions of customer profile features.

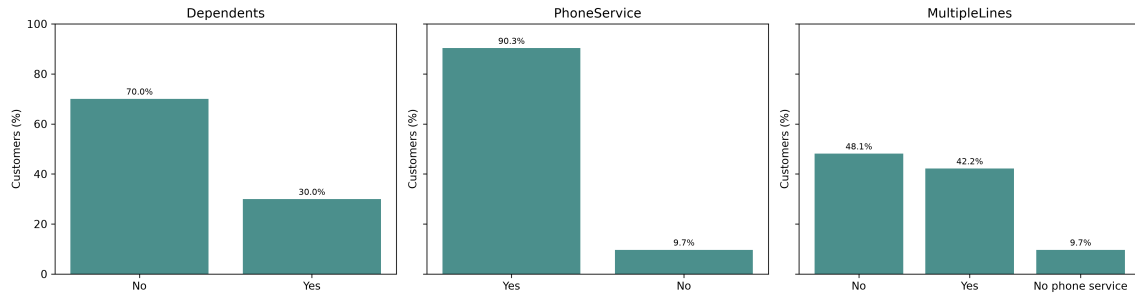


Figure 4: Distributions of household and phone service features.

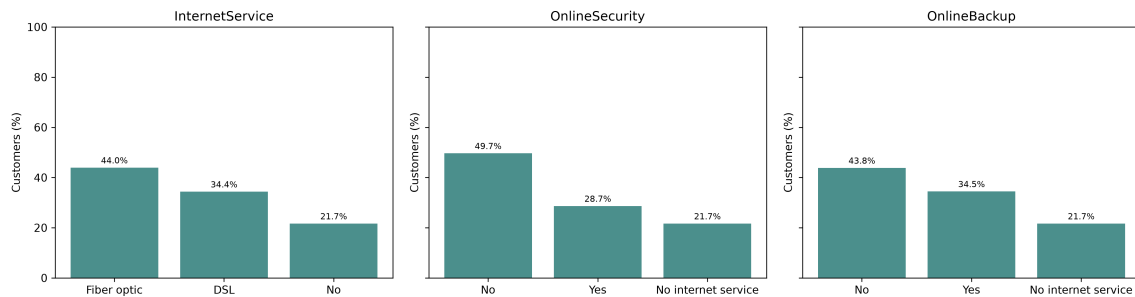


Figure 5: Distributions of internet service and online service features.

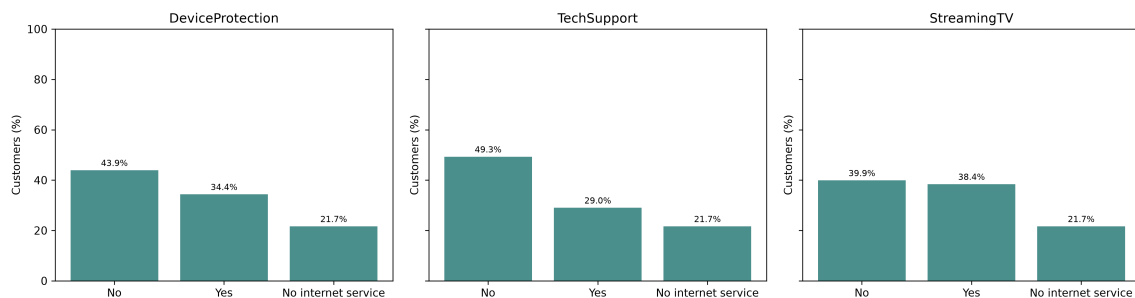


Figure 6: Distributions of device protection, technical support, and streaming TV.

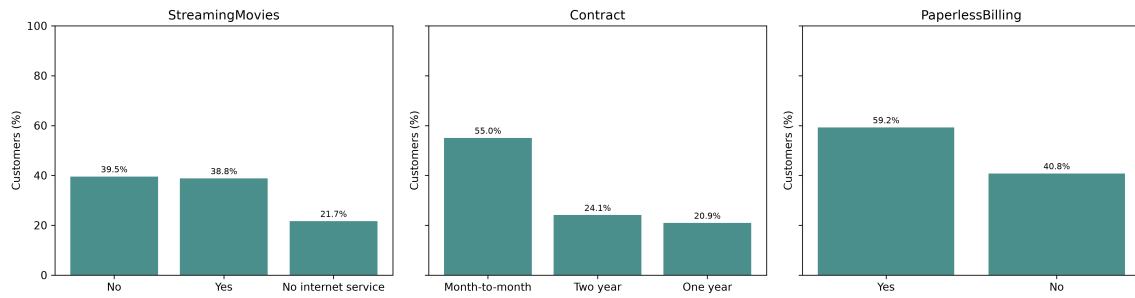


Figure 7: Distributions of streaming movies, contract, and paperless billing.

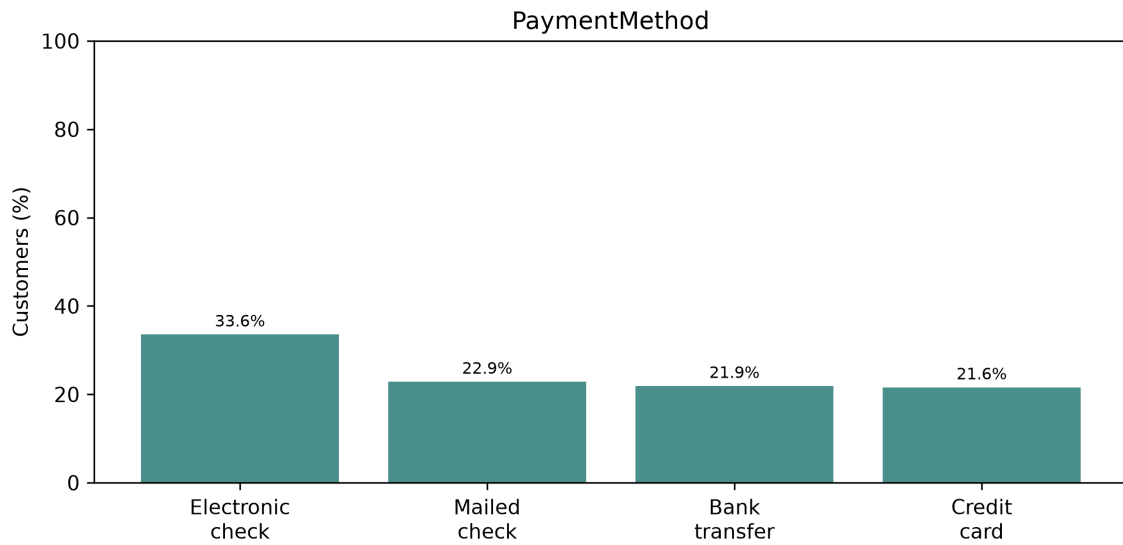


Figure 8: Distribution of customer payment methods.

3.4 Data Quality

The CSV contains no parsed missing cells or duplicate rows, and all 7,043 customer IDs are unique. However, 11 **TotalCharges** values are blank strings (0.16% of records). Is a little portion unusable. There are 7,032 values equal (99.84% of records)

3.5 Feature Relationships

These charts compare each feature with the churn target to identify differences between customers who stayed and those who left. The numerical-feature boxplots compare the distributions of tenure, monthly charges, and total charges across the two target classes. For categorical features, each pair of bars shows the percentage of customers within a category who stayed or churned; the two percentages sum to 100% for each category. These are descriptive associations and do not establish causation.

The numerical comparisons show that churned customers have a shorter median tenure (10 months versus 38) and higher mean monthly charges (74.44 versus 61.27). Among categorical features, churn is more common for month-to-month contracts (42.7%) than one-year (11.3%) or two-year contracts (2.8%), and for fiber-optic internet (41.9%) and electronic-check payments (45.3%).

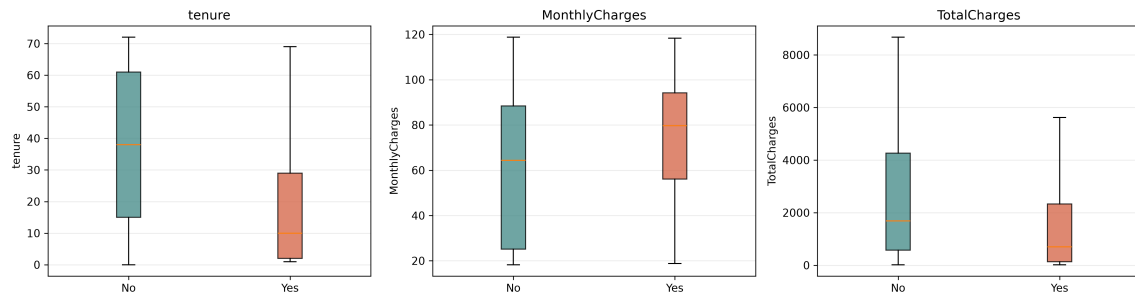


Figure 9: Numeric feature distributions by churn status.

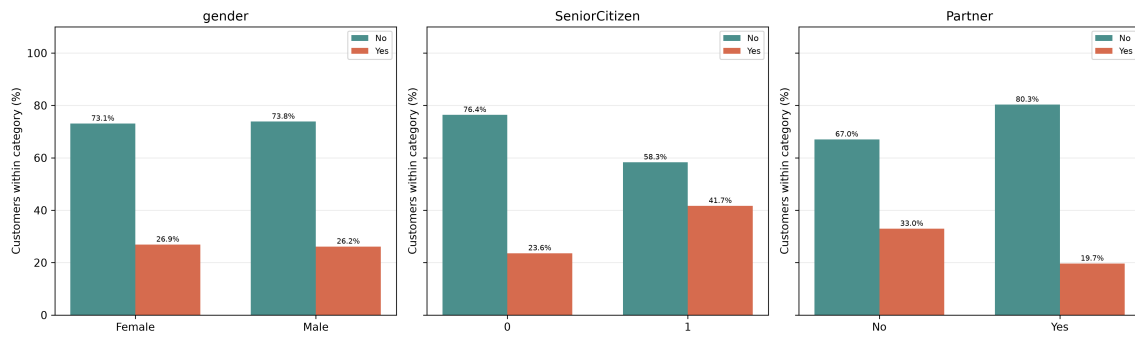


Figure 10: Churn rates by customer profile features.

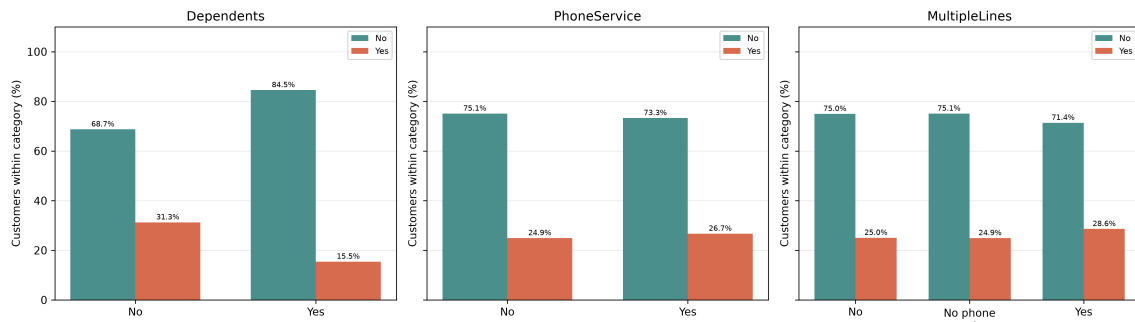


Figure 11: Churn rates by household and phone features.

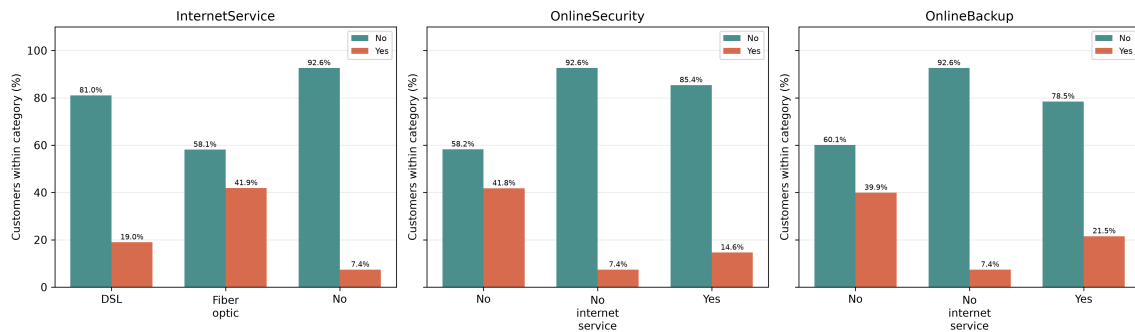


Figure 12: Churn rates by internet and online security features.

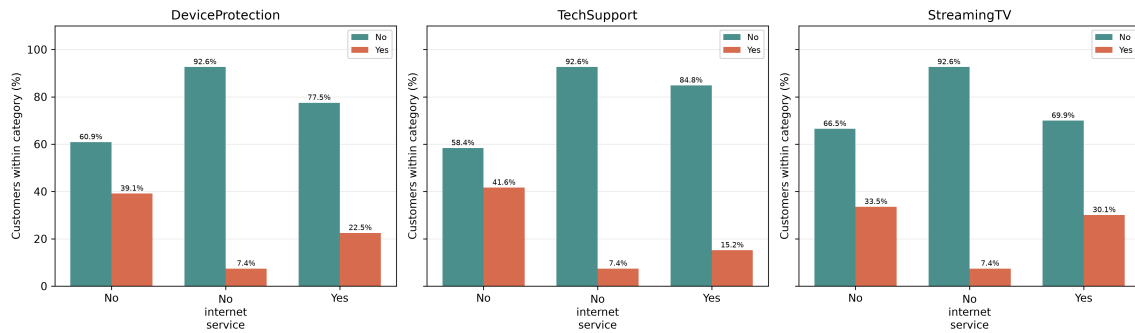


Figure 13: Churn rates by device protection, tech support, and streaming TV.

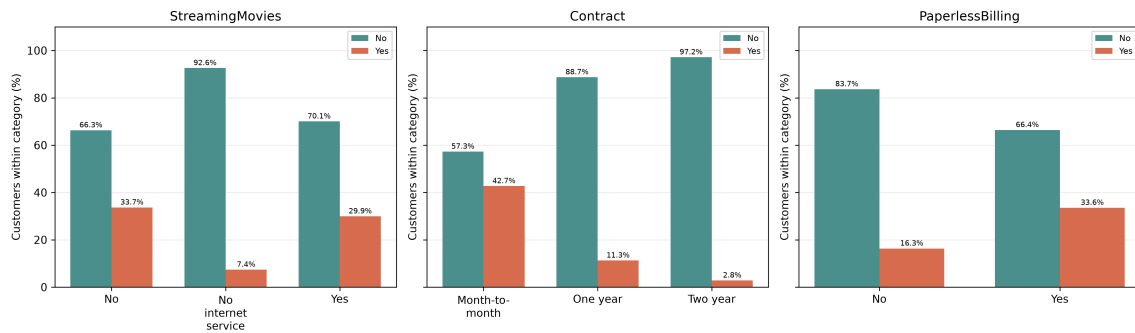


Figure 14: Churn rates by streaming movies, contract, and billing features.

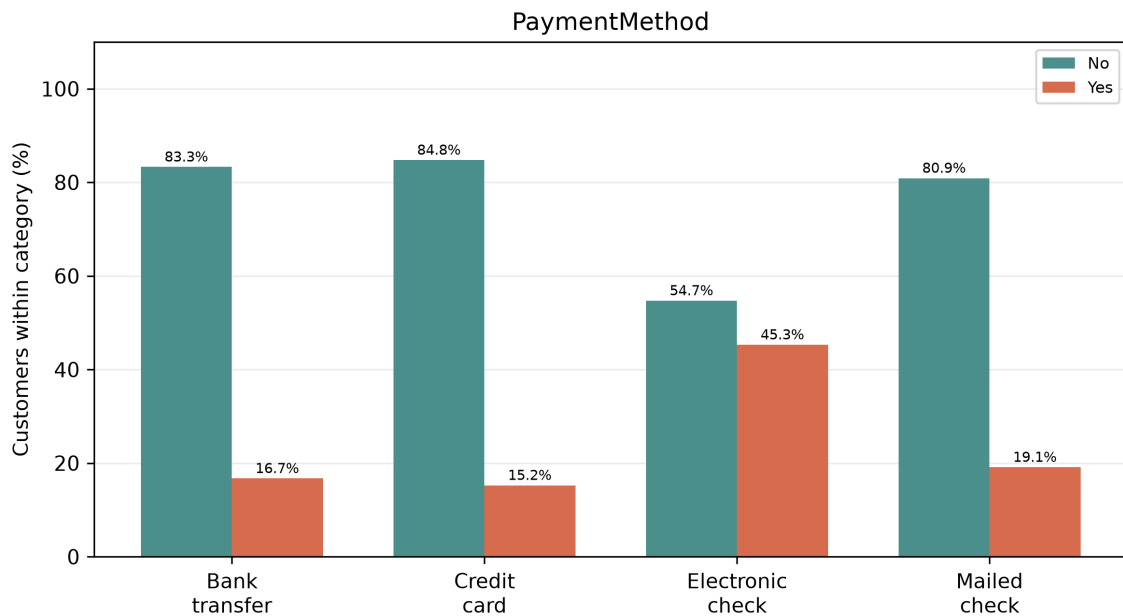


Figure 15: Churn rates by payment method.

4 Preprocessing Pipeline

The preprocessing pipeline is implemented by the preprocessing telco file. Consist in a Class with methods Methods that allow realise the preprocessing. This structure allow separate each step in the

processing of data and is possible add steps simply adding methods to the class. No data augmentation is applied because the nature of the dataset not has images, isn't necessary this technique.

4.1 Format Standardization

The pipeline selects the customer and service features used for modeling and excludes `customerID`, since it uniquely identifies each row and does not describe customer behavior. Categorical inputs remain in their original form until after splitting, when they are encoded using the training feature schema.

4.2 Missing Data Handling

`TotalCharges` is converted to numeric values using imputing. Its 11 blank entries in this dataset belong to customers with zero months of tenure, so they are imputed as zero. The target `Churn` is mapped to binary values, with `No`=0 and `Yes`=1. Any remaining rows with missing values are removed, but in this dataset there isn't that case because the 11 problematic samples where imputed. For the current dataset, all 7,043 rows remain after this step.

4.3 Feature Engineering

Two customer-level features are added. `NewCustomer` is set to 1 when tenure is zero and 0 otherwise. `AvgMonthlyCharges` is calculated as total charges divided by tenure for existing customers; it is set to zero when tenure is zero to avoid division by zero.

4.4 Train, Validation, and Test Split

The data is split before fitting the categorical encoder or scaler. A stratified split with random seed 42 assigns 70% of the records to training and divides the remaining 30% equally between validation and test, yielding approximately 70%/15%/15%. Stratification preserves the target class proportions in each subset. The validation arrays are available as `X_validation` and `y_validation` on the preprocessing object.

4.5 Encoding, Normalization, and Scaling

After splitting, categorical features are converted to one-hot indicators with `OneHotEncoder`. The encoder uses `handle_unknown=ignore` so unseen categories can be transformed using the training feature schema. The encoder and `StandardScaler` are fitted using training data only, then applied unchanged to validation and test data to prevent information leakage. Standard scaling is applied to the continuous features `tenure`, `MonthlyCharges`, `TotalCharges`, and `AvgMonthlyCharges`. The binary `SeniorCitizen` feature and one-hot indicator columns are not scaled.

4.6 Saved Splits

The transformed train, validation, and test arrays, their target labels, and the feature names are saved in compressed NumPy format at `data/telco_preprocessed_splits.npz`. The `preprocess()` method continues to return `X_train`, `X_test`, `y_train`, and `y_test` for compatibility with the existing model notebooks; the validation split is available as attributes and in the saved file.

5 Models

6 Results and Discussion

Explain the experiments you performed to demonstrate that your solution works. Summarize your unit tests in a paragraph, mentioning their philosophy, quantity, and results. Also, discuss any integration and acceptance tests to address software quality and user expectations. If possible, compare your results with other solutions using metrics. Use tables to summarize test definitions and results, and charts (such as box-plots or time series) for comparisons, depending on your project. Highlight the most important findings. Every figure and table must have a descriptive caption and be referenced in the text.

7 Conclusion

Write a couple of paragraphs to summarize your work, results, and achievements. Clearly state why your work was successful and how it helped to solve the problem. End with suggestions for future work or open questions that remain

References

- [1] Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.